

Biofouling effects on ship transport efficiency

Research Project for Environmental Physics
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2025/2026

1 Introduction

In today's maritime world of fast and efficient, but also sustainable ship-travel demands, biofouling can present a huge challenge. It is defined as the accumulation of micro- and macro-organisms on marine vessels and structures (Lindholdt et al., 2015). Extensive accumulation of biofouling on a ship's hull can increase its drag, leading to a reduction in speed but also to increased fuel consumption and hence, the amount of gas emissions (Lindholdt et al., 2015; Puc et al., 2026). Additionally, it can cause the spreading of invasive non-indigenous species across various spatial scales (Puc et al., 2026).

Estimating the drag over a rough surface can be arbitrarily complicated. Specifically, for ship hulls, one can use the boundary layer similarity law (Granville, 1958, 1978), which can be applied to an arbitrarily long hull. However, this requires prior knowledge of the roughness function and assumes that the body is covered with the same roughness everywhere. On the other hand, computational fluid dynamics (CFD) can be employed to predict the hull's resistance with a varying roughness in each grid cell (Song et al., 2019). Several studies have already employed CFD to determine the effects of biofouling on a ship's resistance (see, e.g., Demirel et al., 2017a; Song et al., 2019, and references therein). However, these studies focused mainly on large vessels, specifically, on the the KRISO container ship (KCS). Additionally, despite providing insight in the potential increase in power needed to steer the KCS, they did not provide any analysis on the increased gas emissions. Therefore, it is the goal of this study to estimate the impact of biofouling on smaller recreational boats and to determine its effects on gas emissions. Unlike afore-mentioned studies, which relied on expensive commercial CFD software, we show that for simple 2d flat plate simulations the open-source `OpenFOAM` CFD product (OpenCFD Ltd., 2024) works well.

Different approaches can be undertaken to remedy the effects of biofouling. For example, biocidal coatings can be used preemptively to stall the process of fouling, however, these are hazardous for the environment (Puc et al., 2026). In contrast, the widely used eco-friendlier method of cleaning the hull in dry docks can be very expensive (Puc et al., 2026). Recently, however, Puc et al. (2026) developed a remotely operated vehicle (ROV) within the GreenHull project, which aimed to produce a sustainable method of biofouling removal underwater. Unlike previous methods of in-water cleaning, which released harmful waste into the sea, the GreenHull’s ROV treats the water in three remedying stages, before releasing it back into the open waters (Puc et al., 2026).

In this study, we develop a simple 2d flat plate CFD solver to replicate the hull of a small recreational ship. We then study the effects of biofouling on the ship’s drag, power and fuel consumption and estimate its greenhouse gas emissions. In Sec. 2 we go over the relevant equations and describe the studied system. In Sec. 3 we present the results of the simulations and discuss their implications, whereas in Sec. 4 we discuss the model limitations and give suggestions for future research.

2 Methods

2.1 Relevant theory

Drag forces acting on a ship’s hull

The effective power necessary to move a ship forward is given by

$$P = F_T U, \quad (1)$$

where F_T is the total resistance force and U is the ship’s speed. The former can be decomposed to

$$F_T = F_A + F_R + F_F + F_W, \quad (2)$$

where the terms on the right-hand-side are air resistance, form drag, skin frictional drag and wave-making drag, respectively. Taking into account only hydrodynamical drag forces, the total drag depends on two non-dimensional numbers

$$F_T = (1 + \tilde{k}) F_F(\text{Re}) + F_W(\text{Fr}), \quad (3)$$

namely, the Reynolds number $\text{Re} = UL/\nu$ and Froude number $\text{Fr} = U/\sqrt{gL}$, where L is the ship’s length, ν is the kinematic viscosity and g is the gravitational acceleration. In Eq. (3) we assumed that the form drag is proportional to the frictional resistance through a form factor $\tilde{k}$. This force can be written as

$$F_F(\text{Re}) = \frac{1}{2} \rho C_F(\text{Re}) U^2 S, \quad (4)$$

where ρ is seawater density, C_F is the skin friction coefficient and S is the wetted hull area. Since the wave-making drag force becomes large only for $\text{Fr} \gtrsim 0.2$ (Lindholdt et al., 2015, Fig. 1 therein), we can assume that for $\text{Fr} \leq 0.2$

$$F_T = (1 + \tilde{k})F_F. \quad (5)$$

Boundary layer similarity law

The boundary layer similarity law, developed by Granville (1958, 1978), is the most widely adapted method to estimate the skin friction coefficient over a rough flat plate. In the log law region of the boundary layer, the velocity profile is given by

$$U^+ = \frac{1}{\kappa} \ln(y^+) + B - \Delta U^+(k^+), \quad (6)$$

where $\kappa \approx 0.41$ is the von Karman constant, $y^+ = yU_\tau/\nu$ is the non-dimensional distance from the boundary, $B \sim 5$ is the smooth wall intercept and ΔU^+ is the roughness function, which is dependent on the Reynolds roughness number

$$k^+ = \frac{kU_\tau}{\nu}. \quad (7)$$

Here, k is the roughness length scale and $U_\tau = \sqrt{\tau/\rho}$ is the friction velocity, given for a specific shear stress τ .

For a smooth ship hull, i.e., when $\Delta U^+ = 0$, the C_F is given by the International Towing Tank Conference (ITTC) friction line (ITTC, 2021)

$$C_{F,\text{smooth}} = \frac{0.075}{(\log \text{Re} - 2)^2}. \quad (8)$$

Taking into account Eq. (4), we can write

$$C_F = 2 \left(\frac{U_\tau}{U} \right)^2 = \frac{2}{(U^+)^2}. \quad (9)$$

Thus, the smooth part of the velocity profile, i.e, the first two terms on the right-hand side of Eq. (6), can be rewritten as

$$U_{\text{smooth}}^+ = \frac{1}{\kappa} \ln(y^+) + B = \sqrt{\frac{2}{C_{F,\text{smooth}}}}. \quad (10)$$

The total velocity profile is then simply:

$$U^+ = \sqrt{\frac{2}{C_{F,\text{smooth}}}} - \Delta U^+(k^+(U_\tau)). \quad (11)$$

Note, that this is a transcendent equation since U_τ is a function of U^+ . Therefore, iterative methods have to be used to find the appropriate velocity profile U^+ . In this study, we use the `scipy`'s `root` method within the `optimize` sub-module (Virtanen et al., 2020).

However, the key problem of defining the roughness function remains. These have to be determined empirically, but a general consensus is that they exhibit a logarithmic scaling w.r.t. the Reynolds roughness number. Since we will use the study from Song et al. (2019) (see Sec. 2.2), we will try to replicate their roughness function as best as possible. In the fully turbulent regime ($k^+ \gtrsim 30$), the function is generally given by

$$\Delta U^+ = \frac{1}{\kappa} \ln(1 + k^+) + C, \quad (12)$$

where C is some constant, whereas in the hydraulically smooth region ($k^+ \sim 0$) it is $\Delta U^+ = 0$ (Schultz, 2007). In the intermediate region it is, however, unknown. Therefore, Song et al. (2019) use a regression function fitted to empirical data from Demirel et al. (2017b), which they do not provide. As a roughly similar expression, we choose the function:

$$\Delta U^+ = \begin{cases} \frac{1}{\kappa} \ln\left(\frac{1+k^+}{2.5}\right); & \text{if } k^+ \geq 1.5 \\ 0; & \text{otherwise} \end{cases}, \quad (13)$$

which is also shown on Fig. 1. Note, that a small change in the choice of the roughness function can drastically change the calculated friction coefficient.

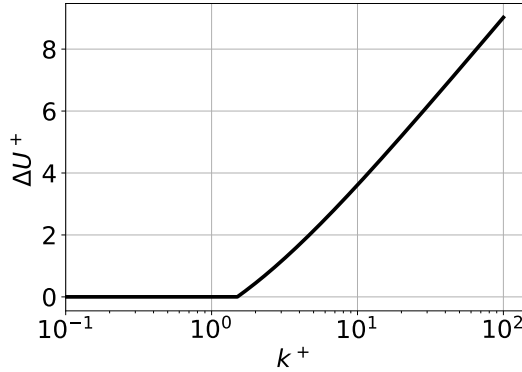


Figure 1: Choice of roughness function for the control experiment.

CFD solver

On the other hand, one can calculate the skin friction coefficient with computational fluid dynamics (CFD). Here, we will use the open-source *Cpp* code `OpenFOAM`

(OpenCFD Ltd., 2024). Specifically, we shall use its steady-state, incompressible and RANS (Reynolds averaged Navier Stokes) solver `simpleFoam`. We will solve the equations for a fluid over a 2-D rough plate, with the `nutkRoughWallFunction` within the k-Omega-SST turbulent scheme. The `nutkRoughWallFunction` provides the boundary condition wall function for the turbulent viscosity based on the turbulent kinetic energy.

Ideally, the computation should be conducted in 3 dimensions and with an actual ship-hull mesh, however, this requires a great amount of computer power and is beyond the scope of this study. Moreover, studies show that the friction coefficient discrepancies between 2-D flat plates, similarity laws and full 3-D ship models aren't that large (Song et al., 2019).

Alas, drag due to wave making, and consequently the total drag, cannot be accounted for without using full 3-D ship models. This means that the difference in power (see Eq. 1) cannot be estimated neither. We will however side step this problem by studying systems where the Froude number $Fr < 0.2$. There, the wave making drag is negligible as discussed at the beginning of this section.

Environmental effects

The total difference in power resulting from biofouling scales as

$$\Delta P = P_{\text{rough}} - P_{\text{smooth}} \propto \Delta C_F. \quad (14)$$

Therefore the relative difference is

$$\delta P = \Delta P / P_{\text{smooth}} = \Delta C_F / C_{F, \text{smooth}}. \quad (15)$$

The mass rate of fuel can be written as (Æsøy et al., 2024)

$$\dot{m} = P \times \text{SFOC}, \quad (16)$$

where SFOC is the specific fuel oil consumption coefficient (in kg / kW h) and where we have denoted the time derivative with a dot. The former is engine-specific and varies depending on the engine load. Finally, converting the mass rate of fuel to the mass rate of greenhouse gas emissions requires a conversion factor ECF, which varies from engine type to the greenhouse gas in question. Therefore, the final difference in the mass rate of released greenhouse gasses is

$$\Delta \dot{m}_{\text{gas}} = \frac{\Delta C_F}{C_{F, \text{smooth}}} P_{\text{smooth}} \text{SFOC} \times \text{ECF}. \quad (17)$$

2.2 Case studies overview

Control

To ensure that our algorithms for the similarity law and the CFD simulation are working correctly we conduct a control experiment, in which we compare the results to the study by Song et al. (2019). The parameters used in the simulations are compiled in Tab. 1. The results are shown in Fig. 2. Notice, that the values are somewhat similar to the control from Song et al. (2019, Fig. 8 therein). The CFD simulation outputs from this study are better matched to their values, compared to the similarity results. This is mostly due to the fact that just a tiny change in the roughness function results in a relatively large difference in the computed output ($\Delta C_F \approx \pm 0.2 \times 10^{-3}$). The smaller observed discrepancies in the CFD simulation could be due to a different kinematic viscosity value, discretization and other parameters. Although, they are most likely explained by the choice of the solver and the simulation length. In Song et al. (2019) they used an unsteady state solver and presumably let the simulation run for longer with a shorter time step considering that they had access to a supercomputer.

Simulation type	Parameter	Description
SL	$\kappa = 0.41$	von Karman constant
SL&CFD	$L = 230 \text{ m}$	ship length taken from (Song et al., 2019)
SL&CFD	$U = 19 \text{ kts}$	ship speed taken from (Song et al., 2019)
SL&CFD	$\nu = 1.1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$	seawater kinematic viscosity estimated from average Adriatic Sea temperature
CFD	$C_S = 0.5$	Roughness constant default value
CFD	$y^+ = 30$	normalized distance from the boundary, chosen as best output match

Table 1: Compilation of parameters used in the control experiment. SL stands for similarity law.

Main study

Now that we have verified that our two algorithms reproduce the friction coefficient results relatively well, we can move on to the main topic of this study, i.e., the friction

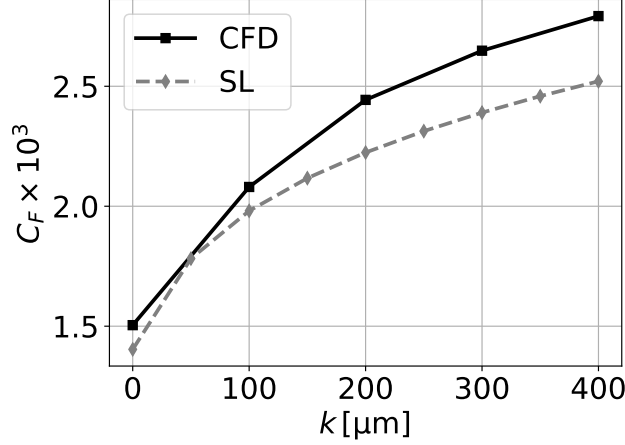


Figure 2: Results of the control experiment. Friction coefficient vs roughness height using the similarity law (SL, dashed grey line) and CFD (solid black line).

coefficient of biofouled smaller recreational boats and its consequences. Later we shall also discuss some effects of biofouling removal with the use of ROVs, specifically the one designed in Puc et al. (2026). There, they reported on the ROV’s efficiency in cleaning the Slovenian tourist ship *Laho* (Wikipedia, 2026), whose specifics we will use for the drag simulation in this study as well.

The ROV is used mainly for the removal of macrofouling. We will thus focus our study on particles of varying heights ranging from 1 – 5 mm. Demirel et al. (2017b, see Eq. 14 therein) provided an empirical regression function for mapping barnacle heights and spatial coverage to the roughness length scale k . Here, we will assume that half of *Laho*’s hull is covered with barnacles. Puc et al. (2026) report that after applying their ROV, 95 % of the barnacles were removed, leading to a final spatial coverage of 2.5 %. However, the regression function in Demirel et al. (2017b) is only valid for spatial-coverages in the range of 10-50 %. The extrapolation to 2.5 % produces unrealistic numbers, therefore, we shall only compare the fully smooth case and the 50 % fouled case. The equivalent roughness length scales are compiled in Tab. 2.

h [mm]	1	2	3	4	5
k [μm]	135	308	534	815	1150

Table 2: Equivalent grain size k calculated from barnacle heights h using the function in Demirel et al. (2017b) with a spatial coverage of 50 %.

The ship *Laho* is 37 m long, with a top speed of 7 kts and is powered by a V-12

Simulation type	Parameter	Description
SL	$\kappa = 0.41$	von Karman constant
SL&CFD	$L = 37 \text{ m}$	ship length taken from (Wikipedia, 2026)
SL&CFD	$U = 4 - 7 \text{ kts}$	ship speed taken from (Wikipedia, 2026)
SL&CFD	$\nu = 1.1 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$	seawater kinematic viscosity estimated from average Adriatic Sea temperature
CFD	$C_S = 0.5$	Roughness constant default value
CFD	$y^+ = 100$	normalized distance from the boundary, chosen as best output match
CFD	$\text{SFOC} = 200 \text{ g kW}^{-1} \text{ h}^{-1}$	specific fuel oil consumption, chosen as average from Æsøy et al. (2024)
CFD	$\text{ECF} = 3.2 \text{ kg of CO}_2/\text{kg of fuel}$	emission conversion factor, taken from Climatiq (2023)
CFD	$\rho = 0.83 \text{ kg l}^{-1}$	marine diesel oil density
CFD	$\rho = 1028 \text{ kg m}^{-3}$	seawater density
CFD	price of oil 1.82 EUR l^{-1}	taken from GlobalPetrolPrices (2026)

Table 3: Compilation of parameters used in the main study. SL stands for similarity law.

diesel engine running on gas oil with 282 kW of power (Wikipedia, [2026](#)). Hence, its Froude number is $\text{Fr} < 0.19$, depending on the ship’s speed. For a marine gas oil engine the emission conversion factors are $\text{ECF} = 3.2 \text{ kg of CO}_2/\text{kg of fuel}$, and $\text{ECF} = 0.03 \text{ g of CH}_4/\text{kg of fuel}$ for CO_2 and CH_4 , respectively ([Climatiq, 2023](#)). The SFOC is engine specific and depends largely on the engine load. Here, we will use the roughly average value of $\text{SFOC} = 200 \text{ g kW}^{-1} \text{ h}^{-1}$ ([Æsøy et al., 2024](#)). The parameters used in this case study are compiled in Tab. [3](#).

Alas, determining the actual operational power of the *Laho*’s engine is tricky as we do not know its efficiency. Hence, we shall leave the total operational power used in the smooth hull case as a free parameter and normalize all the results with

respect to it.

3 Results and discussion

Fig. 3 shows the friction coefficient's dependence on the roughness height k for *Laho*. Clearly, an observable discrepancy arises between the results from the similarity law and the CFD simulations. One cannot comment on which prediction is more accurate as both could be wrong due to the choice of parameters, which are ambiguous in the absence of experimental data. For example, the roughness function from Eq. (13) may not be appropriate for this specific case, therefore, leading to errors in the similarity law calculation. On the other hand, the choice of y^+ and other convergence parameters, which were optimized in order to reduce the large computational times, may lead to unrealistic results in the CFD simulations.

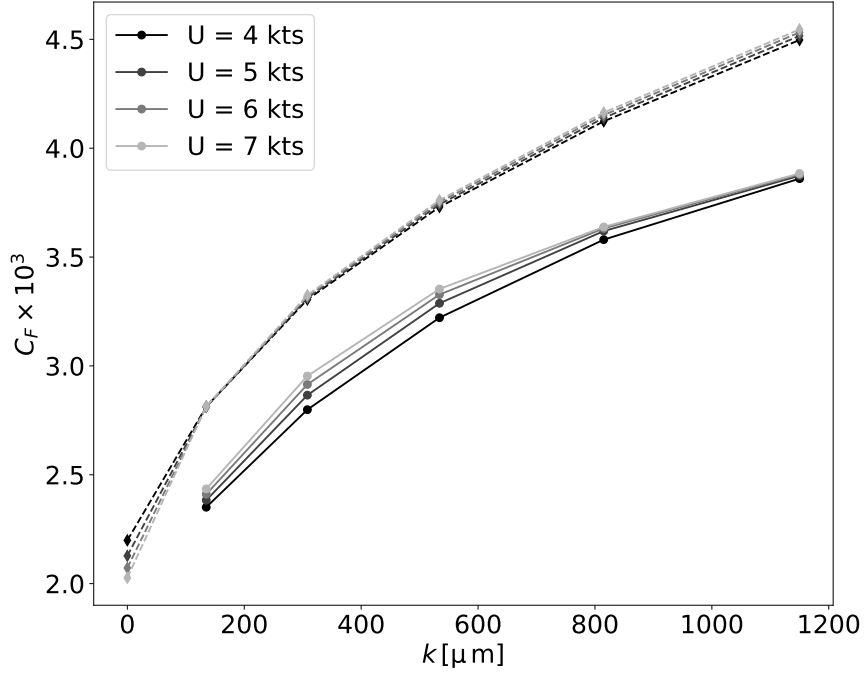


Figure 3: Friction coefficient vs roughness height using the similarity law (dashed lines) and CFD (solid lines) at different speeds of the ship *Laho*.

Nevertheless, a general pattern of an always-slower-increasing function of C_F w.r.t. k is observed. This is equivalent to the dependence from the control experiment. An interesting observation is made when comparing the dependencies using

different cruising speeds. At $k = 0$, a higher speed results in a lower frictional coefficient. This is expected, since in the smooth case C_F is inversely proportional to the Reynolds number (see Eq. 8). In contrast, as k increases, a shift occurs, where the C_F values increase with higher ship speeds U . However, in the fully turbulent regime the dependence on speed vanishes. This is again expected from the similarity law, because at large Roughness Reynolds numbers ($k^+ \gg 1$) one can rewrite Eq. (6) to

$$U^+ = \frac{1}{\kappa} \ln \left(\frac{y}{k} \right) + B - C.$$

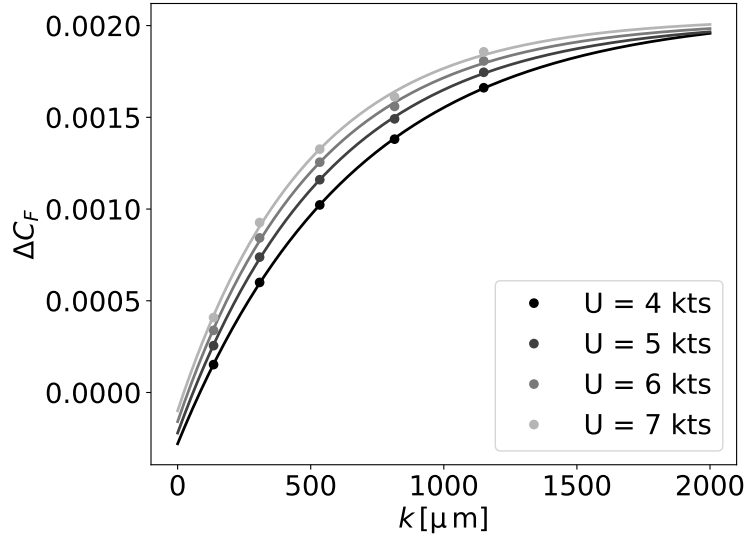


Figure 4: Regression lines on CFD results for the friction coefficient estimation from roughness height.

Hereinafter, we shall use the results obtained from the CFD simulations. We start by estimating the friction coefficient (by means of extrapolation) in the smooth case (at $k = 0$) and compare it to the ITTC values from Fig. 3, which will give us some insight in the potential uncertainty of our estimations. We do this by fitting a regression line

$$\Delta C_F(k) = C_F(k) - C_{F,\text{smooth}} = A \{1 - \exp(b - ck)\}, \quad (18)$$

where A, b, c are model parameters. The regression results are shown on Fig. 4, whereas the regression parameters are compiled in Tab. 4. The regression lines seem to approximate the calculated CFD friction coefficients rather well, however, there is no reason why this particular regression function should be the most appropriate.

On the contrary, a non-constant asymptotic function seems more likely, where the friction coefficient isn't bounded for large k , but rather increases, though be it ever slower. Potential candidates are then the square-root or the logarithmic function. However, I was unable to achieve a good fit with these functions. Calculating ΔC_F at $k = 0$

$$\Delta C_F(k = 0) = A [1 - \exp(b)],$$

and normalizing the result with respect to the smooth value, results in a (5 – 13) % uncertainty estimate.

speed [kts]	parameter	value	uncertainty [%]
4	A	2.1×10^{-3}	–13
	b	0.13	
	c	1.5×10^3	
5	A	2.0×10^{-3}	–10
	b	0.10	
	c	1.8×10^3	
6	A	2.0×10^{-3}	–8
	b	0.08	
	c	1.9×10^3	
7	A	2.0×10^{-3}	–5
	b	0.04	
	c	2.1×10^3	

Table 4: Regression parameters from Eq. (18). The last column shows the uncertainty estimate $\Delta C_F(k = 0)/C_{F,\text{smooth}}$.

Fig. 5 shows how the operational power and CO₂ emissions increase in the different roughness regimes compared to the smooth regime. For example, at the top speed of $U = 7$ kts and for $k = 1150 \mu\text{m}$ (5 mm barnacle height spanning over 50 % of the ship's hull area) the increase in power and emissions are almost 100 % and 0.64 kg CO₂ / kWh, respectively. Recall, that the latter is given in normalized units w.r.t. the operational power in the smooth case. Both quantities increase with higher speeds and larger roughness. However, since power is also a function of speed, specifically $P \propto U^3$ (see Sec. 2), we can also look at the emissions normalized to a transformed surface area $\tilde{S}$ (Fig. 6). Here, $\tilde{S}$ is the surface area divided by 2 (see Eq. 4) and may perhaps include some other engine efficiency conversion factors. The modified normalization shows a much larger gradient in the direction of speed increase as oppose to normalizing the emissions to the operational power. This is an important realization, hence, we will emphasize this note here further. Even if the ship's speed increases slightly the increase in power and hence the emissions can be substantial.

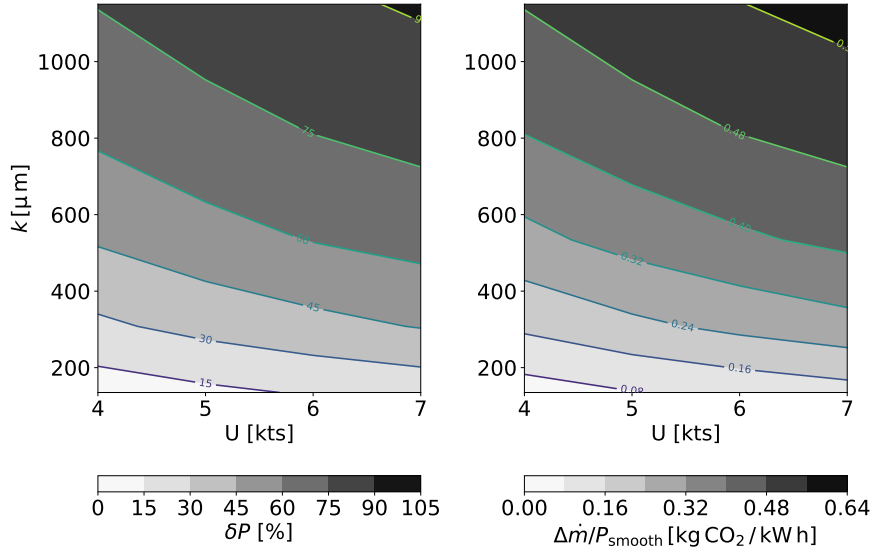


Figure 5: Contour plots of the power (**left**) and CO₂ emission increase (**right**) for different values of ship speed (U) and roughness height (k).

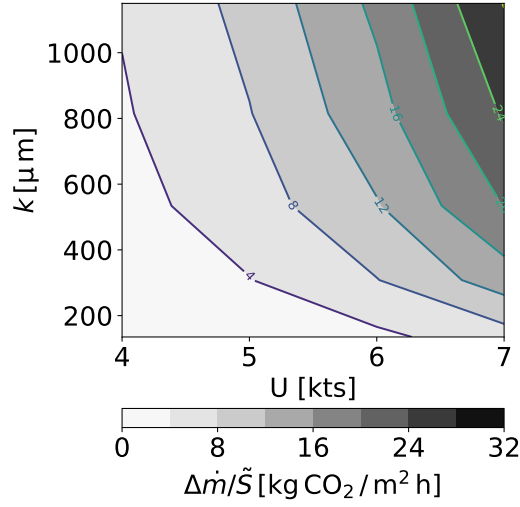


Figure 6: Contour plot of the CO₂ emission increase normalized to the transformed ship surface area $\tilde{S}$ for different values of ship speed (U) and roughness height (k).

This increase can be observed on Fig. 7 for a specific value of roughness height $k = k_{\max} = 1150 \mu\text{m}$. Here, one can see how the increase in CO₂ emissions and

operational costs, both normalized to $\tilde{S}$, scale with the ship's speed. The latter was computed from the mass rate of fuel consumption by converting the mass into the current price of marine diesel oil in Slovenia.

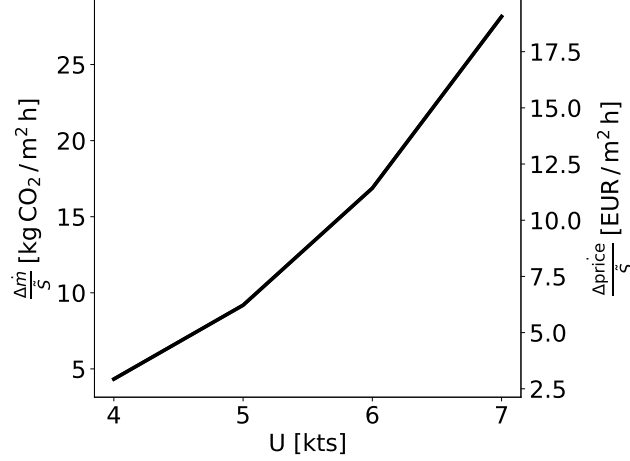


Figure 7: Increase in CO₂ emissions (left-hand axis) and operational-cost (right-hand axis) with respect to ship speed U . Both quantities on the y-axis are normalized to the transformed ship surface area $\tilde{S}$.

4 Limitations and outlook

The models presented here are very simple and certainly prone to large errors. Many of the input parameters are somewhat guessed from combining literature and model-fitting. Ideally, experimental work would have to be conducted to determine the correct roughness function specific for smaller boats like *Laho*. Next, a proper 3-D simulation should be conducted to incorporate varying roughness wall functions, wave-making drag effects and a full-scale ship model. Additionally, this would allow for modeling of the biofouling effect on the ship's propeller, which likely decreases the ship's speed (Puc et al., 2026; Song et al., 2019). Finally, operational characteristics such as the steering power, and the SPOC curve should be measured, to accurately determine the ship's increase in power consumption and greenhouse gas emissions.

Some interesting further studies could include solving optimization problems w.r.t. the ship's emissions, operational cost or journey time. These could include ship cleaning strategies and determine their most efficient and cost-effective real-world application by taking into account biofouling growth and cleaning times.

A Data Availability

The code needed to reproduce the results shown in this study is available online on GitHub at <https://github.com/PerharicC/shipBiofoulingDrag>.

B Acknowledgements

The coding was done in *python* (pre- and post-processing), whereas the openFOAM processing (OpenCFD Ltd., 2024) was done in *c++*. The graphs were created with *python*'s graphing library matplotlib (<https://matplotlib.org/>), whereas the results and parameters were stored using the JSON (<https://www.json.org/json-en.html>), pandas (<https://pandas.pydata.org/>) and pickle (<https://ironpython-test.readthedocs.io/en/latest/library/pickle.html>) libraries. The openFOAM code was analyzed and manipulated with the help of *cursor* (<https://cursor.com/>) – an AI coding agent.

References

- Æsøy, Lene, Houxiang Zhang, and Ann R. Nerheim (2024). “Specific Fuel Oil Consumption Models for Simulating Energy Consumption of Wellboats”. In: *Modeling, Identification and Control* 45.1, pp. 1–14. DOI: [10.4173/mic.2024.1.1](https://doi.org/10.4173/mic.2024.1.1).
- Climatiq (2023). *Emission Factor: Marine gas oil*. Accessed on April 17th. URL: <https://www.climatiq.io/data/emission-factor/8be06433-f0a2-4db3-a095-655ea1711241>.
- Demirel, Yigit Kemal, Osman Turan, and Atilla Incecik (2017a). “Predicting the effect of biofouling on ship resistance using CFD”. In: *Applied Ocean Research* 62, pp. 100–118. ISSN: 0141-1187. DOI: <https://doi.org/10.1016/j.apor.2016.12.003>. URL: <https://www.sciencedirect.com/science/article/pii/S0141118716305685>.
- Demirel, Yigit Kemal et al. (2017b). “Effect of barnacle fouling on ship resistance and powering”. In: *Biofouling* 33.10. PMID: 28980835, pp. 819–834. DOI: [10.1080/08927014.2017.1373279](https://doi.org/10.1080/08927014.2017.1373279). URL: <https://doi.org/10.1080/08927014.2017.1373279>.
- GlobalPetrolPrices (2026). *Slovenia Diesel prices, 13-Apr-2026*. Accessed on April 20th. URL: https://www.globalpetrolprices.com/Slovenia/diesel_prices/#:~:text=The%20current%20price%20of%20diesel%20fuel%20in,based%20on%20the%20latest%20update%20from%2013%2DApr%2D2026.
- Granville, P. S. (1958). “The Frictional Resistance and Turbulent Boundary Layer of Rough Surfaces”. In: *Journal of Ship Research* 2.04, pp. 52–74. ISSN: 0022-4502. DOI: [10.5957/jsr.1958.2.4.52](https://doi.org/10.5957/jsr.1958.2.4.52).

- Granville, P. S. (1978). *Similarity-Law Characterization Methods for Arbitrary Hydrodynamic Roughnesses*. Tech. rep. DAVID W TAYLOR NAVAL SHIP RESEARCH and DEVELOPMENT CENTER BETHESDA MD SHIP PERFORMANCE DEPT. URL: <https://apps.dtic.mil/sti/html/tr/ADA053563/>.
- ITTC (2021). *ITTC Quality System Manual Recommended Procedures and Guidelines; Procedure: Resistance Test*. Tech. rep. International towing tank conference. URL: <https://www.ittc.info/media/11780/75-02-02-01.pdf>.
- Wikipedia (2026). *Laho*. Retrieved on April 13th. URL: <https://sl.wikipedia.org/wiki/Laho>.
- Lindholdt, A. et al. (2015). “Effects of biofouling development on drag forces of hull coatings for ocean-going ships: a review”. In: *Journal of Coatings Technology and Research* 12, pp. 415–444. DOI: [10.1007/s11998-014-9651-2](https://doi.org/10.1007/s11998-014-9651-2).
- OpenCFD Ltd. (2024). Version 2412. URL: <https://www.openfoam.com/>.
- Puc, Uroš et al. (2026). “Development and Field Testing of a Cavitation-Based Robotic Platform for Sustainable In-Water Hull Cleaning”. In: *Journal of Marine Science and Engineering* 14.2. ISSN: 2077-1312. DOI: [10.3390/jmse14020227](https://doi.org/10.3390/jmse14020227). URL: <https://www.mdpi.com/2077-1312/14/2/227>.
- Schultz, Michael P. (2007). “Effects of coating roughness and biofouling on ship resistance and powering”. In: *Biofouling* 23.5. PMID: 17852068, pp. 331–341. DOI: [10.1080/08927010701461974](https://doi.org/10.1080/08927010701461974). URL: <https://doi.org/10.1080/08927010701461974>.
- Song, Soonseok, Yigit Kemal Demirel, and Mehmet Atlar (2019). “An investigation into the effect of biofouling on the ship hydrodynamic characteristics using CFD”. In: *Ocean Engineering* 175, pp. 122–137. ISSN: 0029-8018. DOI: <https://doi.org/10.1016/j.oceaneng.2019.01.056>. URL: <https://www.sciencedirect.com/science/article/pii/S0029801819300460>.
- Virtanen, Pauli et al. (2020). “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python”. In: *Nature Methods* 17, pp. 261–272. DOI: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2).